CP-Direct-KV

Reconstruction-Free CP-Decomposed KV Attention on a Single Device

Born-compressed coefficients, bounded SVD-free adaptation, RoPE-pair-aligned residuals, and GEMM-friendly kernels

CP-Direct-KV represents old layer-wise KV tensors with CANDECOMP/PARAFAC factors and derives attention algebra that reads those factors directly. Its goal is a practical tensor method whose runtime is dominated by regular matrix operations rather than dense-cache reconstruction.

Positioning. This is a *Technique Paper* and the lower-risk companion to CATT-KV. CP is not claimed as a new decomposition. The proposed contributions are the CP layout selected for KV semantics, exact reconstruction-free attention algebra, a two-mode encode/adapt lifecycle, and a latency-aware rank/residual policy validated against low-bit quantization.

Rigorous research proposal • July 2026

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1 Executive decision

One-sentence key idea

Factor each old KV block over token, KV-head, RoPE-pair, and pair-component modes, cache the CP token coefficients and small global/block factors, and evaluate logits and value aggregation by GEMMs over the factors without reconstructing a token's full K or V vector.

CP-Direct-KV trades some of CATT-KV's multiscale expressiveness for simpler execution. Its direct attention equations reduce to dense, contiguous matrix products; therefore it has a more credible path to speedup on Jetson and standard NVIDIA GPUs. Its scientific risk is instead representational: CP can be ill-conditioned, its best low-rank approximation may be unstable, and a shared token factor may require too high a rank on heterogeneous heads. Regularization and explicit stop gates are therefore part of the proposal.

Axis	Required contribution	Gate
Representation	CP over token/head/RoPE-pair/component with separate K and V factors	$\geq 4\times$ physical KV reduction
Online cost	Fixed-factor encoding or at most two block CP-ALS sweeps	Positive payback within 128 future queries
Execution	Direct logits and V aggregation through factors	No $B \times H \times d$ old-cache allocation
Quality	Attention-weighted fitting plus pair-block sparse innovations	Near-full-cache task quality at selected budget
Competitiveness	Same-byte comparison with INT4 and modern tensor/matrix baselines	Better latency or quality at matched memory

2 Research gap and novelty boundary

Existing low-rank KV methods mostly use matrices: Palu factorizes projection weights and caches a latent state [1]; Eigen Attention computes in a reduced subspace [2]; OjaKV adapts a matrix basis online [3]; and RAP prunes RoPE-aligned coordinate pairs to avoid reconstruction [4]. Tensor methods include MPO-based DecoQuant [5] and partial Tucker plus residual allocation in JoLT [6]. Sparse dictionaries also reconstruct KV from selected atoms [7].

CP-Direct-KV occupies a narrower space: it uses a CP model whose factors correspond to the semantic modes of the physical cache and rewrites both sides of attention in factor form. A July 2026 search did not find a prior KV method with this exact CP execution path; that search is evidence for a novelty hypothesis, not proof. A final submission must repeat the search across arXiv, OpenReview, DBLP, Semantic Scholar, and patent databases.

Family	Factorization target	Runtime adaptation	Attention path	Difference
Palu / Eigen	Feature matrix	None / calibration	Reduced projection or reconstruction	No head–pair CP structure
OjaKV	Per-cache matrix basis	Online PCA-style	Reconstructed approximate KV	Matrix update; no CP factor algebra
DecoQuant	Matrix reshaped as MPO	Data-free	Dequantization kernel	Tensorization primarily improves quantization
JoLT	Token/feature Tucker	Block decomposition	Compressed storage focus	Tucker plus randomized-SVD variant
Lexico	Sparse dictionary	Per-token coding	Sparse KV reconstruction	Does not compute softmax attention in factor space
CP-Direct-KV	Token, head, pair, component CP	Encode-only or bounded block ALS	**Direct factor GEMMs**	Joint semantic modes, pair residual, payback gate

Claim allowed only if the experiments support it

Do not claim "the first tensor KV cache." Claim a CP-specific representation/execution co-design only after confirming the novelty search and demonstrating that direct factor attention, rather than CP storage alone, changes the end-to-end Pareto frontier.

3 Thinking template and logic chain

Stage	Proposal content
Research background	Long-context inference is memory-bandwidth bound because every decoded token reads all prior K/V. Low-rank and tensor compression help only when their representation can be created and consumed more cheaply than the dense cache.
Limitation 1	Matrix projections discard multiway head/frequency structure; existing tensor methods do not expose a CP attention algebra optimized for regular GEMMs.
Limitation 2	A purely offline factor basis is brittle under prompt shift, while unconstrained runtime CP-ALS is too expensive and may be unstable.
Limitation 3	CP reconstruction error alone does not protect softmax routing, and arbitrary sparse residuals can destroy kernel regularity.
Key idea	Use a semantic CP cache that is born as token coefficients or adapted once per aged block, then consumed directly with RoPE-pair-aligned sparse corrections.
Challenge 1	Choose CP modes and rank so factors compress both GQA and MHA caches without forcing unrelated K/V structure.
Challenge 2	Bound runtime fitting and make both attention phases factor-executable.
Challenge 3	Correct attention-critical approximation failures without irregular scalar sparsity.
Module A	Separate K/V semantic CP with encode-only and bounded-adapt modes.
Module B	Direct CP logits/value kernels plus payback-aware rank control.
Module C	Query-weighted fitting, RoPE-pair block residuals, and attention-error guarantees.
Contributions	C1 CP cache representation (Sec. 4); C2 reconstruction-free equations and kernels (Sec. 6); C3 bounded adaptation, residual, and analysis (Secs. 5–9); C4 end-to-end evaluation (Sec. 11).

The four consistency checks pass. The representation module answers the missing multiway layout; the lifecycle/direct-kernel module answers runtime feasibility; the structured-residual module answers routing risk. Each claimed contribution is backed by a named method or experiment section.

4 Formal problem and CP representation

For a layer and closed token block, reshape the K cache as

$$\mathcal{K} \in \mathbb{R}^{B\times H\times P\times 2}, \qquad P = d/2,$$

where the final mode preserves the two coordinates of each RoPE pair. Use separate rank-R_K and rank-R_V CP models:

$$\widehat{\mathcal{K}}_{shpu} = \sum_{r=1}^{R_K} A^K_{sr} B^K_{hr} C^K_{pr} D^K_{ur}, \tag{1}$$

$$\widehat{\mathcal{V}}_{shpu} = \sum_{r=1}^{R_V} A^V_{sr} B^V_{hr} C^V_{pr} D^V_{ur}. \tag{2}$$

Here s is token, h is KV head, p is RoPE/frequency pair, and $u \in \{0, 1\}$ is the pair component. K and V never share factors by default; shared factors are an ablation because the two objects play different roles and have different compressibility.

Proposition 1 (CP storage)**.** *The scalar count for one K block is $R_K(B + H + P + 2)$ instead of $2BHP$; V is analogous. The paper reports physical bytes after padding, factor scales, residual indices, and alignment.*

For GQA, H may be small enough that a separate CP per head is competitive. Therefore evaluate two layouts: **joint-head CP** as above and **head-private CP**, $\mathcal{K}_h \in \mathbb{R}^{B\times P\times 2}$. The controller chooses by measured bytes and latency, not by an assumption that joint factorization is always better.

5 Two-mode cache lifecycle

5.1 Mode E: encode-only, born compressed

Offline calibration learns layer-specific non-token atoms $\Phi_r = B_{:r} \circ C_{:r} \circ D_{:r}$. For each generated full K or V token tensor x_s, compute only its coefficient vector

$$a_s = (\Phi^\top \Phi + \gamma I)^{-1} \Phi^\top \operatorname{vec}(x_s),$$

using a precomputed regularized Gram inverse, append a_s to A, then release x_s. This path is SVD-free and makes cache growth R rather than $2HP$ scalars per token. Training the atoms uses a calibration corpus and an attention-distillation loss; the base LLM weights remain frozen.

5.2 Mode A: bounded block adaptation

Encode-only factors may fail under domain shift. In adaptive mode, keep a native exact hot block. When it ages, initialize CP from the calibrated atoms and run at most $J \in \{1, 2\}$ regularized CP-ALS sweeps. Each sweep is implemented through MTTKRP and small Gram solves; no SVD is permitted. Randomized CP-ALS sampling is a fallback for large blocks [8]. Factors are normalized each sweep, component weights are clipped, and nearly collinear components are merged or pruned to limit degeneracy.

The runtime selects Mode E, Mode A, INT4, or native storage from a per-layer cost table. Mode A is admitted only if measured fit cost can be recovered over the predicted remaining queries.

Why both modes belong in one paper

They share the same CP cache and direct kernels. Mode E tests the lowest-overhead deployment point; Mode A tests whether a tiny amount of prompt adaptation buys enough quality to justify its cost. The paper does not propose a general CP optimizer; it proposes a bounded KV lifecycle.

6 Reconstruction-free CP attention

Let a query for KV head h be $q_h \in \mathbb{R}^{P\times 2}$. Define

$$g^K_{hr}(q) = B^K_{hr} \sum_{p=1}^{P} \sum_{u=0}^{1} q_{hpu} C^K_{pr} D^K_{ur}.$$

Then the score for all B tokens is

$$\widehat{s}_h = A^K g^K_h(q)/\sqrt{d}. \tag{3}$$

Equation (3) is a matrix-vector product; it never constructs $\widehat{\mathcal{K}}_{s,h,:,:}$. After online softmax gives $\alpha_h \in \mathbb{R}^B$, define

$$z^V_{hr} = \sum_{s=1}^{B} \alpha_{hs} A^V_{sr}.$$

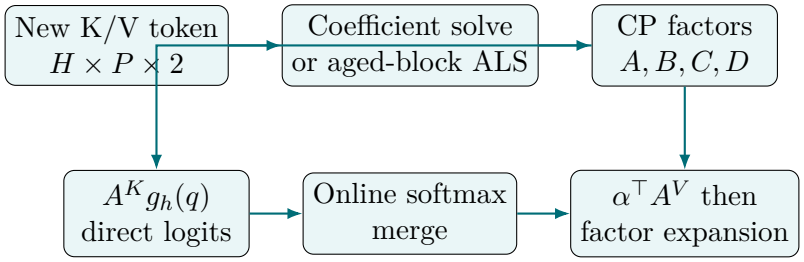
The attention output is

$$\widehat{o}_{hpu} = \sum_{r=1}^{R_V} z^V_{hr} B^V_{hr} C^V_{pr} D^V_{ur}. \tag{4}$$

This is again a short GEMM followed by factor scaling. Equations (3) and (4) are algebraically exact for the approximate CP cache.

Proposition 2 (Direct-execution complexity)**.** *For one head and block, CP logits require $O(R_K d + BR_K)$ operations and CP value aggregation requires $O(BR_V + R_V d)$, versus $O(Bd)$ for each dense phase. CP reduces bytes read when $R \ll d$, but arithmetic is lower only in a rank-dependent region that must be measured.*

The implementation fuses factor scaling, GEMM, sparse residual contributions, and online softmax. The exact hot block and all old CP blocks contribute partial softmax maxima, denominators, and numerators, which are merged with the standard numerically stable block-softmax identity.



7 RoPE handling and structured sparse residual

CP-Direct-KV initially factorizes *post-RoPE* keys. This keeps Equation (3) simple and correct but may raise CP rank because rotation couples token position and frequency pair. The pair mode is retained explicitly so the method can measure and correct that effect rather than hiding it in a flattened feature axis.

Two RoPE variants are evaluated:

1. **Post-RoPE CP**: factorize the cache exactly as consumed by ordinary attention; simplest and primary implementation.

2. **Phase-augmented CP**: add analytic sine/cosine-modulated token factors for selected frequency bands. This increases factor count but may reduce base CP rank. It is an ablation, not a required contribution, because RAP and frequency-aware methods already make generic RoPE claims novelty-sensitive [4].

Let $e^K_{shp:} = K_{shp:} - \widehat{K}_{shp:}$ and similarly for V. The residual unit is an entire two-coordinate RoPE pair. Pair score

$$I_{shp} = \max_{q \in \mathcal{Q}} |q^\top_{hp:} e^K_{shp:}| + \eta \bar{\alpha}_{hs} \left\| e^V_{shp:} \right\|_2$$

selects the top k pair blocks. This avoids scalar index scatter, preserves pair rotation, and targets errors capable of changing attention rather than merely the largest tensor entries.

8 Attention-aware fitting and controller

Calibration and bounded adaptation minimize

$$L = \mathbb{E}_{(K,V,Q)} \left[\left\| \mathrm{softmax}(QK^\top/\sqrt{d})V - \mathrm{softmax}(Q\widehat{K}^\top/\sqrt{d})\widehat{V} \right\|_F^2 \right] \tag{5}$$

$$+ \lambda_K \left\| K - \widehat{K} \right\|^2_{W_K} + \lambda_V \left\| V - \widehat{V} \right\|^2_{W_V} + \gamma \sum_r \left(\|B_{:r}\|_2^2 + \|C_{:r}\|_2^2 + \|D_{:r}\|_2^2 \right), \tag{6}$$

where weighted reconstruction supplies a cheap ALS-compatible surrogate and attention distillation is evaluated periodically. Runtime Mode A uses the surrogate plus one final attention-error check; it does not backpropagate through long sequences during decoding.

For each layer/head layout, the controller solves a small profiled choice:

$$\min_{m \in \{\text{native, INT4, E, A}\}, R, k} \widehat{L}_{\text{attn}} + \lambda M + \mu T_{\text{query}} + \nu T_{\text{fit}}$$

subject to memory and expected-payback constraints. Rank is searched over a small discrete set; no claim of solving CP rank globally is made.

9 Theory targets

Target theorem 1 (Direct CP attention equivalence)**.** *Prove Equations (3)–(4) equal standard softmax attention evaluated on the reconstructed CP tensors, up to floating-point order, while requiring no reconstructed KV allocation. Extend the proof to stable block-softmax merging and sparse pair corrections.*

Target theorem 2 (One-query output error)**.** *For* $\epsilon_K = \max_s \left\| K_s - \widehat{K}_s \right\|_2$, $\epsilon_V = \max_s \left\| V_s - \widehat{V}_s \right\|_2$, *and bounded approximate values, establish*

$$\|o(q) - \widehat{o}(q)\|_2 \le \epsilon_V + 2V_{\max} \|q\|_2 \epsilon_K / \sqrt{d},$$

then tighten it using observed attention mass and the exact sparse pair support.

Proposition 3 (Encode stability)**.** *For fixed atoms* Φ *and ridge* $\gamma > 0$*, the coefficient map* $x \mapsto (\Phi^\top\Phi + \gamma I)^{-1}\Phi^\top x$ *is Lipschitz with constant bounded by the regularized smallest singular value of* Φ*. The implementation monitors the Gram condition number and rejects/merges unstable components.*

Proposition 4 (Payback)**.** *If a block costs* T_c *to encode/adapt and saves* $\Delta T > 0$ *per later query, it breaks even after* $\lceil T_c/\Delta T \rceil$ *queries. The controller stores native or INT4 KV whenever* $\Delta T \le 0$ *or predicted lifetime is shorter.*

These results do not prove that low CP rank exists. Compressibility is an empirical property, and worst-case KV compression can be impossible [9].

10 Implementation design

The reference implementation first validates direct equations against explicit CP reconstruction to 10^{-4} relative error in FP32. Production kernels use contiguous factor layouts: A token-major, B head-major, and fused $C \odot D$ pair atoms. Key execution becomes one query–atom GEMV followed

by A GEMV; value execution reverses the path. Batched requests and heads promote these GEMVs to GEMMs.

Implementation stages:

1. PyTorch/TensorLy CP fitting and explicit/direct equality tests.
2. Mode-E coefficient encoder with cached Gram inverse and condition monitor.
3. One/two-sweep regularized CP-ALS using MTTKRP for Mode A.
4. CUDA kernels for direct K logits, online softmax merge, V aggregation, and pair residuals.
5. Hugging Face cache integration, then Jetson Orin and a second NVIDIA platform.

11 Experimental plan

11.1 Research questions

1. Which layout—joint-head or head-private—has the best rank/latency frontier for GQA and MHA?
2. Does Mode E remain accurate across domains; when does one-sweep Mode A recover its failures?
3. Does attention-weighted fitting outperform Frobenius CP-ALS at equal physical bytes?
4. Are pair-block residuals more accurate and faster than scalar or token residuals?
5. At what (B, R, H, d) does direct CP attention beat dense and INT4 kernels after amortization?

11.2 Models and workloads

Develop on LLaMA-3.2-1B/3B, then include at least two families and both GQA/MHA if compute permits. Use PG19 perplexity, RULER, LongBench, a reasoning benchmark, and controlled distribution shifts (code, dialogue, scientific text, and multilingual prompts). Test 4K–32K context where the base model is valid, batch 1 for edge latency, and larger batches on the second GPU.

11.3 Baselines

Use full BF16/FP16, native INT8/INT4, KIVI [11], Palu, Eigen Attention, OjaKV, RAP, DecoQuant, JoLT, Lexico, CP storage with dense reconstruction, and CATT-KV when available. A recent preprint argues quantization can dominate rank reduction at matched bytes [10]; therefore CP-Direct-KV cannot be evaluated only against SVD or Tucker.

11.4 Metrics and mandatory ablations

Measure actual allocated KV bytes, metadata, bytes read/token, encoding and adaptation time, payback queries, TTFT, median/P95 inter-token latency, tokens/s, energy/token, OOM context, perplexity, task accuracy, attention-output error, and Gram condition number. Include failed and rerouted blocks in controller statistics.

Ablate: K versus V rank; joint-head versus head-private; flattened feature versus pair/component modes; Mode E versus Mode A; ALS sweeps; calibration size/domain; attention objective; residual granularity; post-RoPE versus phase augmentation; factor precision; direct versus reconstructed attention; and controller fallback.

12 Reviewer attacks and pre-registered failure cases

Falsification gate

Pivot if CP rank required for near-lossless quality removes the byte advantage, if Gram conditioning requires frequent component resets, or if direct CP remains slower than INT4 beyond 16K context. In that case, retain the direct-attention analysis as a negative/diagnostic result and move to CATT-KV or a quantized CP dictionary only if a new Pareto point is demonstrated.

- **"CP is classical."** Correct; novelty is the semantic layout, lifecycle, direct attention algebra, and measured systems frontier. Never claim decomposition novelty.
- **"This is just a Kronecker dictionary."** Show the CP tensor interpretation, mode-wise rank/ablation, and why direct K/V equations exploit those factors. Acknowledge the connection.
- **"Offline atoms fail under shift."** Report cross-domain degradation and the exact fraction/cost of Mode-A adaptation or fallback.
- **"CP can be ill-posed."** Normalize, regularize, monitor conditioning, cap sweeps, and publish failure statistics.
- **"Memory savings do not imply speed."** Present end-to-end latency including encoding, adaptation, residuals, and fallback; make payback a first-class result.

13 Paper outline and contribution discipline

1. **Introduction:** memory-bandwidth problem, why reconstruction defeats compression, CP direct execution idea.
2. **Diagnosis/related work:** strongest matrix, tensor, sparse, and quantized KV methods; naive CP study.
3. **Representation/lifecycle:** semantic CP, Mode E, Mode A, stability controls.
4. **Direct attention:** key/value equations, block-softmax, pair residual, CUDA mapping.
5. **Objective/analysis:** attention weighting, error and cost bounds, payback controller.
6. **Evaluation:** quality/memory/latency frontier, distribution shift, ablations, failures.
7. **Limitations:** CP degeneracy, fixed-atom bias, custom kernels, architecture dependence.

Limit the paper to three method contributions plus evaluation. Do not add collaborative inference, multi-device scheduling, a benchmark, or universal tensor theory to the first submission.

14 Execution schedule

Weeks	Deliverable	Exit criterion
1–2	Recorded-KV CP pilot	Rank, conditioning, and direct-vs-dense microbenchmark curves
3–4	Reference direct attention	Explicit/direct outputs agree; storage accounting complete
5–7	Mode E and Mode A	Encode cost, one-sweep adaptation benefit, payback table
8–10	CUDA kernels	No dense old-KV allocation and positive speed region
11–14	Model integration/evaluation	Two model families, strong quantization/tensor baselines
15–18	Analysis and writing	Verified proofs, artifact, complete draft

15 Final verdict

Claim allowed only if the experiments support it

If the gates pass, the defensible thesis is: *a semantic CP cache converts long-context K and V reads into regular factor GEMMs, and a bounded encode/adapt lifecycle plus pair-aligned residuals preserves attention while ensuring compression is used only when it repays its own cost.*

CP-Direct-KV is the more feasible of the two tensor projects and the better first implementation. Its novelty ceiling is lower, so a top paper requires decisive systems evidence and a clear attention-aware method, not merely CP factors. Target MLSys or ACM ASPLOS when kernels/payback lead; target NeurIPS/ICLR only if the attention objective, stability analysis, and cross-model results are substantive; target ACM MobiSys if energy and sustained single-device operation become the headline.

References

[1] C.-C. Chang et al. *Palu: KV-Cache Compression with Low-Rank Projection.* ICLR 2025. https://openreview.net/forum?id=LWMS4pk2vK

[2] U. Saxena et al. *Eigen Attention: Attention in Low-Rank Space for KV Cache Compression.* Findings of EMNLP 2024. https://aclanthology.org/2024.findings-emnlp.899/

[3] Y. Zhu et al. *OjaKV: Context-Aware Online Low-Rank KV Cache Compression.* Findings of ACL 2026. https://aclanthology.org/2026.findings-acl.494/

[4] J. Xin et al. *RAP: KV-Cache Compression via RoPE-Aligned Pruning.* arXiv:2602.02599, 2026. https://arxiv.org/abs/2602.02599

[5] P. Liu et al. *Unlocking Data-free Low-bit Quantization with Matrix Decomposition for KV Cache Compression.* ACL 2024. https://aclanthology.org/2024.acl-long.133/

[6] R. Krishnan and V. Schulz. *A JoLT for the KV Cache: Near-Lossless KV Cache Compression via Joint Tucker and JL-Residual Allocation for LLMs.* arXiv:2607.12550, 2026. https://arxiv.org/abs/2607.12550

[7] J. Kim et al. *Lexico: Extreme KV Cache Compression via Sparse Coding over Universal Dictionaries.* ICML 2025. https://arxiv.org/abs/2412.08890

[8] C. Battaglino, G. Ballard, and T. G. Kolda. *A Practical Randomized CP Tensor Decomposition.* SIAM J. Matrix Analysis and Applications, 2018. https://arxiv.org/abs/1701.06600

[9] L. Haverbeck et al. *The Risk of KV Cache Compression.* arXiv:2607.01520, 2026. https://arxiv.org/abs/2607.01520

[10] S. Salfati. *Quantization Dominates Rank Reduction for KV-Cache Compression.* arXiv:2604.11501, 2026. https://arxiv.org/abs/2604.11501

[11] Z. Liu et al. *KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache.* ICML 2024. https://arxiv.org/abs/2402.02750